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Semiconductor device structure and methods of forming the same

Abstract

Embodiments of the present disclosure relates to a method for forming a semiconductor device structure. The method includes including forming one or more conductive features in a first interlayer dielectric (ILD), forming an etch stop layer on the first ILD, forming a second ILD over the etch stop layer, forming one or more openings through the second ILD and the etch stop layer to expose a top surface of the one or more first conductive features, wherein the one or more openings are formed by a first etch process in a first process chamber, exposing the one or more openings to a second etch process in a second process chamber so that the shape of the or more openings is elongated, and filling the one or more openings with a conductive material.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application Ser. No. 63/414,538 filed Oct. 9, 2022, which is incorporated by reference in its entirety.

BACKGROUND
(1) As the semiconductor industry has progressed into nanometer technology process nodes in pursuit of higher device density, higher performance, and lower costs, challenges from both fabrication and design issues have resulted in the development of three-dimensional designs, such as a Fin Field Effect Transistor (FinFET). FinFET devices typically include semiconductor fins with high aspect ratios and in which channel and source/drain regions are formed. A gate is formed over and along the sides of the fin structure (e.g., wrapping) utilizing the advantage of the increased surface area of the channel to produce faster, more reliable, and better-controlled semiconductor transistor devices. However, with the decreasing in scaling, the critical dimension (CD) between adjacent contact features becomes smaller. To achieve high performance of integrated circuits (ICs), the higher resistance due to the smaller CD to neighboring metal features in the bank-end-of-line (BEOL) has become a critical issue.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIGS. **1-4** are perspective views of various stages of manufacturing a semiconductor device structure, in accordance with some embodiments.
(3) FIGS. **5**A-**20**A are cross-sectional side views of various stages of manufacturing the semiconductor device structure of FIG. **4** taken along cross-section A-A, in accordance with some embodiments.
(4) FIGS. **5**B-**20**B are cross-sectional side views of various stages of manufacturing the semiconductor device structure of FIG. **4** taken along cross-section B-B, in accordance with some embodiments.
(5) FIG. **17**B-**1** is an enlarged view of a portion of the semiconductor device structure being etched in accordance with some embodiments.
(6) FIG. **18-1** illustrates a top view of a portion of the third ILD showing the profile of the via contact opening prior to the reactive ion beam etch process.
(7) FIG. **18-2** illustrates a top view of a portion of the third ILD showing the profile of the via contact openings after the reactive ion beam etch process in accordance with some embodiments.
DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(9) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “on,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(10) FIGS. **1-20**B illustrate various stages of manufacturing a semiconductor device structure **100** in accordance with various embodiments of this disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. **1-20**B and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable.
(11) FIGS. **1-4** are perspective views of various stages of manufacturing a semiconductor device structure **100**, in accordance with some embodiments. In FIG. **1**, a first semiconductor layer **104** is formed on a substrate **102**. The substrate may be a part of a chip in a wafer. In some embodiments, the substrate **102** is a bulk semiconductor substrate, such as a semiconductor wafer. For example, the substrate **102** is a silicon wafer. The substrate **102** may include silicon or another elementary semiconductor material such as germanium. In some other embodiments, the substrate **102** includes a compound semiconductor. The compound semiconductor may include gallium arsenide, silicon carbide, indium arsenide, indium phosphide, another suitable semiconductor material, or a combination thereof. In some embodiments, the substrate **102** is a semiconductor-on-insulator (SOI) substrate. The SOI substrate may be fabricated using a separation by implantation of oxygen (SIMOX) process, a wafer bonding process, another applicable method, or a combination thereof.
(12) The substrate **102** may be doped with P-type or N-type impurities. As shown in FIG. **1**, the substrate **102** has a P-type region **102**P and an N-type region **102**N adjacent to the P-type region **102**P, and the P-type region **102**P and N-type region **102**N belong to a continuous substrate **102**, in accordance with some embodiments. In some embodiments of the present disclosure, the P-type region **102**P is used to form a PMOS device thereon, whereas the N-type region **102**N is used to form an NMOS device thereon. In some embodiments, an N-well region **103**N and a P-well region **103**P are formed in the substrate **102**, as shown in FIG. **1**. For example, the N-well region **103**N may be formed in the substrate **102** in the P-type region **102**P, whereas the P-well region **103**P may be formed in the substrate **102** in the N-type region **102**N. The P-well region **103**P and the N-well region **103**N may be formed by any suitable technique, for example, by separate ion implantation processes in some embodiments. By using two different implantation mask layers (not shown), the P-well region **103**P and the N-well region **103**N can be sequentially formed in different ion implantation processes.
(13) The first semiconductor layer **104** is deposited over the substrate **102**, as shown in FIG. **1**. The first semiconductor layer **104** may be made of any suitable semiconductor material, such as silicon, germanium, III-V semiconductor material, or combinations thereof. In one exemplary embodiment,

the first semiconductor layer **104** is made of silicon. The first semiconductor layer **104** may be formed by an epitaxial growth process, such as metal-organic chemical vapor deposition (MOCVD), metal-organic vapor phase epitaxy (MOVPE), plasma-enhanced chemical vapor deposition (PECVD), remote plasma chemical vapor deposition (RP-CVD), molecular beam epitaxy (MBE), hydride vapor phase epitaxy (HYPE), liquid phase epitaxy (LPE), chloride vapor phase epitaxy (Cl-VPE), or any other suitable process.

(14) In FIG. **2**, the portion of the first semiconductor layer **104** disposed over the N-well region **103**N is removed, and a second semiconductor layer **106** is formed over the N-well region **103**N and adjacent the portion of the first semiconductor layer **104** disposed over the P-well region **103**P. A patterned mask layer (not shown) may be first formed on the portion of the first semiconductor layer **104** disposed over the P-well region **103**P, and the portion of the first semiconductor layer **104** disposed over the N-well region **103**N may be exposed. A removal process, such as a dry etch, wet etch, or a combination thereof, may be performed to remove the portion of the first semiconductor layer **104** disposed over the N-well region **103**N, and the N-well region **103**N may be exposed. The removal process does not substantially affect the mask layer (not shown) formed on the portion of the first semiconductor layer **104** disposed over the P-well region **103**P, which protects the portion of the first semiconductor layer **104** disposed over the P-well region **103**P. Next, the second semiconductor layer **106** is formed on the exposed N-well region **103**N. The second semiconductor layer **106** may be made of any suitable semiconductor material, such as silicon, germanium, III-V semiconductor material, or combinations thereof. In one exemplary embodiment, the second semiconductor layer **106** is made of silicon germanium. The second semiconductor layer **106** may be formed by the same process as the first semiconductor layer **104**. For example, the second semiconductor layer **106** may be formed on the exposed N-well region **103**N by an epitaxial growth process, which does not form the second semiconductor layer **106** on the mask layer (not shown) disposed on the first semiconductor layer **104**. As a result, the first semiconductor layer **104** is disposed over the P-well region **103**P in the N-type region **102**N, and the second semiconductor layer **106** is disposed over the N-well region **103**N in the P-type region **102**P.

(15) Portions of the first semiconductor layer **104** may serve as channels in the subsequently formed NMOS device in the N-type region **102**N. Portions of the second semiconductor layer **106** may serve as channels in the subsequently formed PMOS device in the P-type region **102**P. In some embodiments, the NMOS device and the PMOS device are FinFETs. While embodiments described in this disclosure are described in the context of FinFETs, implementations of some aspects of the present disclosure may be used in other processes and/or in other devices, such as planar FETs, dual-gate FETs, tri-gate FETS, nanosheet channel FETs, forksheet FETs, Horizontal Gate All Around (HGAA) FETs, Vertical Gate All Around (VGAA) FETs, complementary FETs, negative-capacitance FETs, and other suitable devices.

(16) In FIG. **3**, a plurality of fins **108***a*, **108***b*, **110***a*, **110***b* are formed from the first and second semiconductor layers **104**, **106**, respectively, and STI regions **121** are formed. The fins **108***a*, **108***b*, **110***a*, **110***b* may be patterned by any suitable method. For example, the fins **108***a*, **108***b*, **110***a*, **110***b* may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer (not shown) is formed over a substrate and patterned using a photolithography process. Spacers (not shown) are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the substrate and form the fins.

(17) The fins **108***a*, **108***b* may each include the first semiconductor layer **104**, and a portion of the first semiconductor layer **104** may serve as an NMOS channel. Each fin **108***a*, **108***b* may also

include the P-well region **103**P. Likewise, the fins **110***a*, **110***b* may each include the second semiconductor layer **106**, and a portion of the second semiconductor layer **106** may serve as a PMOS channel. Each fin **110***a*, **110***b* may also include the N-well region **103**N. A mask (not shown) may be formed on the first and second semiconductor layers **104**, **106**, and may remain on the fins **108***a-b* and **110***a-b*.

(18) Once the fins **108***a-b*, **110***a-b* are formed, an insulating material **112** is formed between adjacent fins **108***a-b*, **110***a-b*. The insulating material **112** may be first formed between adjacent fins **108***a-b*, **110***a-b* and over the fins **108***a-b*, **110***a-b*, so the fins **108***a-b*, **110***a-b* are embedded in the insulating material **112**. A planarization process, such as a chemical-mechanical polishing (CMP) process may be performed to expose the top of the fins **108***a-b*, **110***a-b*. In some embodiments, the planarization process exposes the top of the mask (not shown) disposed on the fins **108***a-b* and **110***a-b*. The insulating material **112** are then recessed by removing a portion of the insulating material **112** located on both sides of each fin **108***a-b*, **110***a-b*. The insulating material **112** may be recessed by any suitable removal process, such as dry etch or wet etch that selectively removes the insulating material **112** but does not substantially affect the semiconductor materials of the fins **108***a-b*, **110***a-b*. The insulating material **112** may include an oxygen-containing material, such as silicon oxide, carbon or nitrogen doped oxide, or fluorine-doped silicate glass (FSG); a nitrogen-containing material, such as silicon nitride, silicon oxynitride (SiON), SiOCN, SiCN; a low-K dielectric material (e.g., a material having a K value lower than that of silicon dioxide); or any suitable dielectric material. The insulating material **112** may be formed by any suitable method, such as low-pressure chemical vapor deposition (LPCVD), plasma enhanced CVD (PECVD) or flowable CVD (FCVD). The insulating material **112** may be shallow trench isolation (STI) region, and is referred to as STI region **121** in this disclosure.

(19) In some alternative embodiments, instead of forming first and second semiconductor layers **104**, **106** over the substrate **102**, the fins **108***a-b*, **110***a-b* may be formed by first forming isolation regions (e.g., STI regions **121**) on a bulk substrate (e.g., substrate **102**). The formation of the STI regions may include etching the bulk substrate to form trenches, and filling the trenches with a dielectric material to form the STI regions. The portions of the substrate between neighboring STI regions form the fins. The top surfaces of the fins and the top surfaces of the STI regions may be substantially level with each other by a CMP process. After the STI regions are formed, at least top portions of, or substantially entireties of, the fins are removed. Accordingly, recesses are formed between STI regions. The bottom surfaces of the STI regions may be level with, higher, or lower than the bottom surfaces of the STI regions. An epitaxy is then performed to separately grow first and second semiconductor layers (e.g., first and second semiconductor layers **104**, **106**) in the recesses created as a result of removal of the portions of the fins, thereby forming fins (e.g., fins **108***a-b*, **110***a-b*). A CMP is then performed until the top surfaces of the fins and the top surfaces of the STI regions are substantially co-planar. In some embodiments, after the epitaxy and the CMP, an implantation process is performed to define well regions (e.g., P-well region **103**P and N-well region **103**N) in the substrate. Alternatively, the fins are in-situ doped with impurities (e.g., dopants having P-type or N-type conductivity) during the epitaxy. Thereafter, the STI regions are recessed so that fins of first and second semiconductor layers (e.g., fins **108***a-b*, **110***a-b*) are extending upwardly over the STI regions from the substrate, in a similar fashion as shown in FIG. **3**.

(20) In some alternative embodiments, one of the fins **108***a-b* (e.g., fin **108***a*) in the N-type region **102**N is formed of the second semiconductor layer **106**, and the other fin **108***b* in the N-type region **102**N is formed of the first semiconductor layer **104**. In such cases, the subsequent S/D epitaxial features **152** formed on the fins **108***a* and **108***b* in the N-type region **102**N may be Si or SiP; the subsequent S/D epitaxial features **152** formed on the fins **110***a* and **110***b* in the P-type region **102**P may be SiGe. In some alternative embodiments, the fins **108***a-b* and **110***a-b* are formed directly from a bulk substrate (e.g., substrate **102**), which may be doped with P-type or N-type impurities to form well regions (e.g., P-well region **103**P and N-well region **103**N). In such cases, the fins are

formed of the same material as the substrate **102**. In one exemplary embodiment, the fins and the substrate **102** are formed of silicon.
(21) In FIG. **4**, one or more sacrificial gate stacks **128** are formed on a portion of the fins **108***a-b*, **110***a-b*. Each sacrificial gate stack **128** may include a sacrificial gate dielectric layer **130**, a sacrificial gate electrode layer **132**, and a mask structure **134**. The sacrificial gate dielectric layer **130** may include one or more layers of dielectric material, such as SiO.sub.2, SiN, a high-K dielectric material, and/or other suitable dielectric material. In some embodiments, the sacrificial gate dielectric layer **130** may be deposited by a CVD process, an ALD process, a PVD process, or other suitable process. The sacrificial gate electrode layer **132** may include polycrystalline silicon (polysilicon). The mask structure **134** may include an oxygen-containing layer and a nitrogen-containing layer. In some embodiments, the sacrificial gate electrode layer **132** and the mask structure **134** are formed by various processes such as layer deposition, for example, CVD (including both LPCVD and PECVD), PVD, ALD, thermal oxidation, e-beam evaporation, or other suitable deposition techniques, or combinations thereof.
(22) The sacrificial gate stacks **128** may be formed by first depositing blanket layers of the sacrificial gate dielectric layer **130**, the sacrificial gate electrode layer **132**, and the mask structure **134**, followed by pattern and etch processes. For example, the pattern process includes a lithography process (e.g., photolithography or e-beam lithography) which may further include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, photoresist developing, rinsing, drying (e.g., spin-drying and/or hard baking), other suitable lithography techniques, and/or combinations thereof. In some embodiments, the etch process may include dry etch, wet etch, other etch methods, and/or combinations thereof. By patterning the sacrificial gate stacks **128**, the fins **108***a-b*, **110***a-b* are partially exposed on opposite sides of the sacrificial gate stacks **128**. While two sacrificial gate stacks **128** are shown in FIG. **4**, it can be appreciated that they are for illustrative purpose only and any number of the sacrificial gate stacks **128** may be formed.
(23) FIGS. **5**A-**20**A are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100** of FIG. **4** taken along cross-section A-A, in accordance with some embodiments. FIGS. **5**B-**20**B are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100** of FIG. **4** taken along cross-section B-B, in accordance with some embodiments. Cross-section B-B is in a plane of the fin **110***b* along the X direction. Cross-section A-A is in a plane perpendicular to cross-section B-B and is in the S/D epitaxial features **152** (FIG. **6**A) along the Y-direction.
(24) In FIGS. **5**A-**5**B, a gate spacer **140** is formed on the sacrificial gate structures **128** and the exposed portions of the first and second semiconductor layers **104**, **106**. The gate spacer **140** may be conformally deposited on the exposed surfaces of the semiconductor device structure **100**. The conformal gate spacer **140** may be formed by ALD or any suitable processes. An anisotropic etch is then performed on the gate spacer **140** using, for example, reactive ion etching (RIE). During the anisotropic etch process, most of the gate spacer **140** is removed from horizontal surfaces, such as tops of the sacrificial gate structures **128** and tops of the fins **108***a-b*, **110***a-b*, leaving the gate spacer **140** on the vertical surfaces, such as on opposite sidewalls of the sacrificial gate structures **128**. The gate spacers **140** may partially remain on opposite sidewalls of the fins **108***a-b*, **110***a-b*, as shown in FIG. **5**A. In some embodiments, the gate spacers **140** formed on the source/drain regions of the fins **108***a-b*, **110***a-b* are fully removed.
(25) The gate spacer **140** may be made of a dielectric material such as silicon oxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), silicon carbide (SiC), silicon oxynitride (SiON), silicon carbon-nitride (SiCN), silicon oxycarbide (SiOC), silicon oxycarbonitride (SiOCN), air gap, and/or any combinations thereof. In some embodiments, the gate spacer **140** include one or more layers of the dielectric material discussed herein.
(26) In FIGS. **6**A-**6**B, the first and second semiconductor layers **104**, **106** of the fins **108***a-b*, **110***a-b*

not covered by the sacrificial gate structures **128** and the gate spacers **140** are recessed, and source/drain (S/D) epitaxial features **152** are formed. For N-channel FETs, the epitaxial S/D features **152** may include one or more layers of Si, SiP, SiC, SiCP, or a group III-V material (InP, GaAs, AlAs, InAs, InAlAs, InGaAs). In some embodiments, the epitaxial S/D features **152** may be doped with N-type dopants, such as phosphorus (P), arsenic (As), etc, for N-type devices. For P-channel FETs, the epitaxial S/D features **152** may include one or more layers of Si, SiGe, SiGeB, Ge, or a group III-V material (InSb, GaSb, InGaSb). In some embodiments, the epitaxial S/D features **152** may be doped with P-type dopants, such as boron (B). The epitaxial S/D features **152** may grow both vertically and horizontally to form facets, which may correspond to crystalline planes of the material used for the substrate **102**. The epitaxial S/D features **152** may be formed by an epitaxial growth method using CVD, ALD or MBE.
(27) In some embodiments, the portions of the first semiconductor layer **104** on both sides of each sacrificial gate structure **128** are completely removed, and the S/D epitaxial features **152** are formed on the P-well region **103**P of the fins **108**_a_-_b_. The S/D epitaxial features **152** may grow both vertically and horizontally to form facets, which may correspond to crystalline planes of the material used for the substrate **102**. In some embodiments, the S/D epitaxial features **152** of the fins **108**_a_-**108**_b_ and **110**_a_-**110**_b_ are merged, as shown in FIG. **6**A. The S/D epitaxial features **152** may each have a top surface at a level higher than a top surface of the first semiconductor layer **104**, as shown in FIG. **6**B.
(28) In FIGS. **7**A-**7**B, a contact etch stop layer (CESL) **160** is conformally formed on the exposed surfaces of the semiconductor device structure **100**. The CESL **160** covers the sidewalls of the sacrificial gate structures **128**, the insulating material **112**, and the S/D epitaxial features **152**. The CESL **160** may include an oxygen-containing material or a nitrogen-containing material, such as silicon nitride, silicon carbon nitride, silicon oxynitride, carbon nitride, silicon oxide, silicon carbon oxide, or the like, or a combination thereof, and may be formed by CVD, PECVD, ALD, or any suitable deposition technique. Next, a first interlayer dielectric (ILD) **162** is formed on the CESL **160**. The materials for the first ILD **162** may include compounds comprising Si, O, C, and/or H, such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, silicon oxide, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. The first ILD **162** may be deposited by a PECVD process or FCVD process or other suitable deposition technique. In some embodiments, after formation of the first ILD **162**, the semiconductor device structure **100** may be subject to a thermal process to anneal the first ILD **162**. After formation of the first ILD **162**, a planarization process is performed to expose the sacrificial gate electrode layer **132**. The planarization process may be any suitable process, such as a CMP process. The planarization process removes portions of the first ILD **162** and the CESL **160** disposed on the sacrificial gate structures **128**. The planarization process may also remove the mask structure **134**.
(29) In FIGS. **8**A-**8**B, the mask structure **134** (if not removed during previous CMP process), the sacrificial gate electrode layers **132** (FIG. **7**B), and the sacrificial gate dielectric layers **130** (FIG. **7**B) are removed. The sacrificial gate electrode layers **132** and the sacrificial gate dielectric layers **130** may be removed by one or more etch processes, such as dry etch process, wet etch process, or a combination thereof. The one or more etch processes selectively remove the sacrificial gate electrode layers **132** and the sacrificial gate dielectric layers **130** without substantially affects the gate spacer **140**, the CESL **160**, and the first ILD **162**. The removal of the sacrificial gate electrode layers **132** and the sacrificial gate dielectric layers **130** exposes a top portion of the first and second semiconductor layers **104**, **106** (only first semiconductor layers **104** can be seen in FIG. **8**A) in the channel region.
(30) In FIGS. **9**A-**9**B, replacement gate structures **177** are formed. The replacement gate structure **177** may include a gate dielectric layer **166** and a gate electrode layer **167** formed on the gate dielectric layer **166**. The gate dielectric layer **166** may include one or more dielectric layers and

may include the same material(s) as the sacrificial gate dielectric layer **130**.

(31) In some embodiments, the gate dielectric layer **166** is a high-K dielectric material such as metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, or any combination thereof. For example, the gate dielectric layer **166** may include hafnium oxide (HfO.sub.2), hafnium silicon oxide (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HMO), hafnium zirconium oxide (HfZrO), lanthanum oxide (LaO), zirconium oxide (ZrO), titanium oxide (TiO), tantalum oxide (Ta.sub.2O.sub.5), yttrium oxide (Y.sub.2O.sub.3), strontium titanium oxide (SrTiO.sub.3), barium titanium oxide (BaTiO.sub.3), barium zirconium oxide (BaZrO), hafnium lanthanum oxide (HfLaO), lanthanum silicon oxide (LaSiO), aluminum silicon oxide (AlSiO), aluminum oxide (Al.sub.2O.sub.3), silicon nitride (Si.sub.3N.sub.4), oxynitrides (SiON), and combinations thereof. In alternative embodiments, the gate dielectric layer **166** may have a multilayer structure such as one layer of silicon oxide (e.g., interfacial layer) and another layer of high-K material. In some embodiments, the gate dielectric layer **166** may be deposited by one or more ALD processes or other suitable processes.

(32) Depending on the application and/or conductivity type of the devices in the N-type region **102**N and the P-type region **102**P, the gate electrode layer **167** may include one or more layers of electrically conductive material, such as polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, AlTi, AlTiO, AlTiC, AlTiN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or any combinations thereof. For devices in the N-type region **102**N, the gate electrode layer **167** may be AlTiO, AlTiC, or a combination thereof. For devices in the P-type region **102**P, the gate electrode layer **167** may be AlTiO, AlTiC, AlTiN, or a combination thereof. The gate electrode layers **167** may be formed by PVD, CVD, ALD, electro-plating, or other suitable method.

(33) In FIGS. **10**A-**10**B, a metal gate etching back (MGEB) process is performed to remove portions of the gate dielectric layer **166** and the gate electrode layer **167**. Recesses **175** are formed in the region between neighboring gate spacers **140** as a result of the removal of the portions of the gate dielectric layer **166** and the gate electrode layer **167**. The recesses **175** are defined by the exposed sidewalls of the gate spacers **140** and the recessed top surfaces of the gate electrode layers **167** and the gate dielectric layers **166**, respectively. The recesses **175** allow for subsequent first dielectric cap layer **141** (FIG. **11**B) to be formed therein and protect the replacement gate structures **177**. The MGEB process may include one or more etching processes, which may be dry etching, wet etching, atomic layer etching (ALE), plasma etching, any suitable etching back, or a combination thereof. The one or more etching processes performed in the MGEB process are selective to materials of the replacement gate structures **177** with respect to the gate spacers **140** and the first ILD **162** so that the top surfaces of the gate electrode layers **167** and the gate dielectric layers **166**, respectively, are at a level lower than top surfaces of the gate spacers **140** and the first ILD **162**.

(34) In FIGS. **11**A-**11**B, a dielectric cap layer **141** is formed in the trenches **175** (FIG. **10**B), over the replacement gate structures **177**. The dielectric cap layer **141** fills in the trenches **175** and over the first ILD **162** to a pre-determined height using a deposition process, such as CVD, PECVD, or FCVD or any suitable deposition technique. A CMP process is then performed to remove excess deposition of the dielectric cap layer **141** outside the trenches **175** until the top surface of the first ILD **162** is exposed. The top surfaces of the first ILD **162**, the CESL **160**, the dielectric cap layer **141**, and the gate spacers **140** are substantially coplanar. The dielectric cap layer **141** defines self-aligned contact (SAC) regions and thus serve as an etch stop layer during subsequent trench and via patterning for metal contacts. The dielectric cap layer **141** can be formed of any dielectric material that has different etch selectivity than the gate spacers **140**, the CESL **160**, and the first ILD **162**. In some embodiments, the dielectric cap layer **141** may include or be formed of an oxygen-containing

material, a nitrogen-containing material, or a silicon-containing material. Exemplary materials for the dielectric cap layer **141** may include, but are not limited to, SiO, HfSi, SiOC, AlO, ZrSi, AlON, ZrO, HfO, TiO, ZrAlO, ZnO, TaO, LaO, TaCN, SiN, SiOCN, Si, SiOCN, ZrN, SiCN, or any combinations thereof.

(35) In FIGS. **12**A-**12**B, a second ILD **176** is formed over the semiconductor device structure **100**. The second ILD **176** may include the same material as the first ILD **162** and be deposited using the same fashion as the first ILD **162**, as discussed above with respect to FIGS. **7**A and **7**B.

(36) In FIGS. **13**A-**13**B, portions of the first ILD **162** and the CESL **160** disposed on both sides of the replacement gate structures **177** are removed. The removal of the portions of the first ILD **162** and the CESL **160** forms a contact opening **146** exposing the S/D epitaxial features **152**. In some embodiments, the upper portion of the exposed S/D epitaxial features **152** is removed. Portions of the first ILD **162** and the dielectric cap layer **141** over the replacement gate structures **177** are also removed. The removal of the portions of the first ILD **162** and the dielectric cap layer **141** forms a contact opening **147** exposing gate electrode layer **167**. In some embodiments, the upper portion of the exposed gate electrode layer **167** is removed. The portions of the first ILD **162**, the CESL **160**, and dielectric cap layer **141** may be removed by one or more etch processes, such as a wet etch, dry etch, or combination thereof. In one embodiment, a first etch process may be performed to form the contact openings **146**, and a second etch process may be performed to form the contact openings **147**. The etchants used by the first etch process removes the first and second ILD **162**, **176** and the CESL **160** without substantially affecting the dielectric cap layer **141**. The etchants used by the second etch process removes the first ILD **162** and the dielectric cap layer **141** without substantially affecting the gate electrode layer **167** and the exposed S/D epitaxial features **152**.

(37) In FIGS. **14**A-**14**B, conductive features **172** and conductive features **173** are then formed over the S/D epitaxial features **152** and the replacement gate structures **177** in the contact openings **146** and **147**, respectively. The conductive features **172** may serve as S/D contacts while the conductive features **173** may serve as gate contacts. The conductive features **172**, **173** may include an electrically conductive material, such as one or more of Ru, Mo, Co, Ni. W, Ti, Ta, Cu, Al, TiN and TaN. The conductive feature **172** may be formed by any suitable process, such as PVD, CVD, ALD, electrochemical plating, or other suitable method. A silicide layer **170** may be formed between the S/D epitaxial feature **152** and the conductive feature **172**. The silicide layer **170** conductively couples the S/D epitaxial features **152** to the conductive feature **172**. The silicide layer **170** is a metal or metal alloy silicide, and the metal may include a noble metal, a refractory metal, a rare earth metal, alloys thereof, or combinations thereof. Once the conductive features **172**, **173** are formed, a planarization process, such as CMP, is performed on the semiconductor device structure **100** until the top surface of the second ILD **176** is exposed.

(38) In FIGS. **15**A-**15**B, an etch stop layer **145** and a third ILD **178** are sequentially formed over the semiconductor device structure **100**. The etch stop layer **145** may be silicon nitride, silicon carbide, silicon oxide, low-k dielectrics such as carbon doped oxides, extremely low-k dielectrics such as porous carbon doped silicon dioxide, the like, or a combination thereof, and deposited by CVD, PVD, ALD, a spin-on-dielectric process, the like, or a combination thereof. In one embodiment, the etch stop layer **145** is silicon nitride. The third ILD **178** may include the same material as the first ILD **162** and be deposited using the same fashion as the first ILD **162**, as discussed above with respect to FIGS. **7**A and **7**B. In some embodiments, the etch stop layer **145** may have a thickness in a range of about 100 Angstroms to about 300 Angstroms, and the third ILD **178** may have a thickness in a range of about 200 Angstroms to about 700 Angstroms.

(39) In FIGS. **16**A-**16**B, portions of the third ILD **178** and the etch stop layer **145** disposed over the conductive features **172**, **173** are removed. The removal of the portions of the third ILD **178** and the etch stop layer **145** forms via contact openings **149** exposing the conductive features **172**, **173**. A patterned layer (not shown) may be first formed on portions the third ILD **178**. The patterned layer has openings at locations aligned with the S/D epitaxial features **152** and the gate electrode

layer **167**. The removal of the portions of the third ILD **178** and the etch stop layer **145** may be performed, using the patterned layer as a mask, by one or more etch processes, such as a wet etch, dry etch, or a combination thereof (i.e., first etch process). In one embodiment, the portions of the third ILD **178** and the etch stop layer **145** are removed using a dry etch process, such as RIE or other suitable anisotropic etch process. Exemplary dry etch process for removing portions of the third ILD **178** and the etch stop layer **145** may utilize a capacitively coupled plasma (CCP) source, an inductively coupled plasma (ICP) source, or glow discharge plasma (GDP) source driven by an RF power generator or a microwave plasma source using a tunable frequency ranging from about 2 MHz to about 2.45 GHz, such as about 13.56 MHz. The chamber may be operated at a pressure in a range of about 0.3 mTorr to about 20 Torr and a temperature of about −80 degrees Celsius to about 240 degrees Celsius. The RF power generator is operated to provide source power between about 100 W to about 2000 W, and the output of the RF power generator controlled by an optional pulse signal having a duty cycle in a range of about 10% to about 90%. An RF biasing power to the substrate support on which semiconductor device structure **100** is disposed may be in a range of about 100 W to about 1200 W. The source power and the biasing power may be controlled so that the ion acceleration energy is between about 20 eV to about 200 eV (e.g., 50 eV to about 150 eV). The dry etch process may use a plasma formed from a gas mixture containing one or more etching gases, such as hydrogen bromide (HBr), chlorine (Cl_2), hydrogen (H_2), methane (CH_4), nitrogen (N_2), helium (He), neon (Ne), krypton (Kr), tetrafluoromethane (CF_4), trifluoromethane (CHF_3), methyl fluoride (CH_3F), difluoromethane (CH_2F_2), hexafluoroethane (C_2F_6), octofluorocyclobutane (C_4F_8), hexafluorobutadiene (C_4F_6), sulfur hexafluoride (SF_6), nitrogen trifluoride (NF_3), HCl (hydrogen chloride), BCl_3 (boron trichloride), oxygen (O_2), other suitable etching gas, and any combinations thereof. An inert gas, such as argon (Ar), may be provided with the etchants to increase bombardment effect and thus, enhanced etch rates of the third ILD **178** and the etch stop layer **145**. In cases where the gas mixture comprises a chlorine-containing gas (e.g., Cl_2), an oxygen-containing gas (e.g., O_2), and argon, the chlorine-containing gas, the oxygen-containing gas, and argon may be introduced into the process chamber at a ratio (Cl_2:O_2:Ar) of about 10:1:5 to about 50:1:5, for example about 20:1:5.

(40) The sidewalls of the via contact openings **149** may be vertical or slanted. In some embodiments, the via contact openings **149** have a sidewall profile in which the dimension at the top is greater than the dimension at the bottom of the via contact openings **149**. The via contact openings **149** may be a round-shaped and/or oval-shaped opening when viewing from top, such as the via contact openings **149** shown in FIG. **18-1**. For example, in some embodiments the via contact openings **149** over the S/D epitaxial features **152** and the gate electrode layer **167** may have a round-shaped profile when viewing from top. In some embodiments, the via contact openings **149** over the S/D epitaxial features **152** and the gate electrode layer **167** may have an oval-shaped profile when viewing from top. In some embodiments, the via contact openings **149** over the S/D epitaxial features **152** may have a round-shaped profile when viewing from top, and the via contact openings **149** over the gate electrode layer **167** may have an oval-shaped profile when viewing from top, or vice versa. Upon formation of the via contact openings **149**, the patterned layer may be removed using any suitable process, such as an ash process. Alternatively, the patterned layer may not be removed until the subsequent second etch process is complete.

(41) In FIGS. **17**A-**17**B, the via contact openings **149** are further etched by a second etch process. In some embodiments, the second etch process is performed in a process chamber separated from the process chamber for performing the first etch process. The second etch process is performed so that at least one dimension of the via contact openings **149** (e.g., X direction or Y direction) is further extended. The second etch process extends, for example, the dimension along the Y direction of the via contact openings **149**, which increases the contact area between the via contact openings **149** and the neighboring metal features in the bank-end-of-line (BEOL). The increase of

the contact area of the via contact openings **149** enable better gap-filling while reducing the contact resistance effectively. In some embodiments, the second etch process is a dry etch process, such as an ion beam etch (IBE), reactive ion beam etch (RIBE), bombardment etch, sputtering etch, or the like. In one exemplary embodiment shown in FIGS. **17**A and **17**B, the second etch process is a reactive ion beam etch. The reactive ion beam etch process may involve forming a plasma **154** and directing a reactive ion beam **155** to the semiconductor device structure **100**. The reactive ion beam **155** may be extracted from the plasma **154** through an extraction aperture **179**, between extraction part **153***a* and extraction part **153***b*, to carry out material removal processes on the third ILD **178** and the etch stop layer **145**. During operation, the semiconductor device structure **100** is moved relative to the reactive ion beam **155**. The reactive ion beam **155** is a beam of charged particles (i.e., ions) to be directed to scan over the semiconductor device structure **100** along the X or Y direction, thereby removing portions of the third ILD **178** and the etch stop layer **145**. After the second etch process, the via contact openings **149** has at least one dimension extended and become via contact openings **149**′. In some embodiments, the shape of the via contact openings **149**′ is extended along the Y direction only. If the via contact openings **149** have a rounded shape before the second etch process, the via contact openings **149**′ may become an elongated circle along the Y-direction or oval in shape. If the via contact openings **149** have an oval shape, the via contact openings **149**′ may become an elongated oval shape along the Y-direction, as to be discussed below with respect to FIGS. **18-1** and **18-2**.
(42) The reactive ion beam etch process may use an inert gas (e.g., He, Ne, Kr, Ar) and/or an etching gas. Suitable etching gas may be a fluorine-containing gas and/or a chlorine-containing gas, such as CF.sub.4, CHF.sub.3, CH.sub.3F, CH.sub.2F.sub.2, C.sub.4F.sub.8, C.sub.4F.sub.6, SF.sub.6, C.sub.2F.sub.6, NF.sub.3, HCl, Bcl.sub.3, and other suitable reactive gases, and any combinations thereof. In some embodiments, an oxygen-containing gas (e.g., O.sub.2, O.sub.3, etc.) may also be used. During etching and/or by directing etch ions in an IBE or RIBE etch process towards the semiconductor device structure **100**, a bias power is also applied to the semiconductor device structure **100** to enhance anisotropic etch of the materials of the third ILD **178** and the etch stop layer **145**. Exemplary reactive ion beam etch process for removing portions of the third ILD **178** and the etch stop layer **145** may utilize a CCP, ICP, or GDP source driven by an RF power generator or a microwave plasma source using a tunable frequency ranging from about 2 MHz to about 2.45 GHz, such as about 13.56 MHz. The chamber may be operated at a pressure in a range of about 0.3 mTorr to about 20 Torr and a temperature of about −80 degrees Celsius to about 240 degrees Celsius. The RF power generator is operated to provide source power between about 100 W to about 1000 W, and the output of the RF power generator controlled by an optional pulse signal having a duty cycle in a range of about 10% to about 90%. The substrate support on which semiconductor device structure **100** is disposed may be biased with respect to the plasma in a range of about 0.1 kV to about 12 kV. The source power and the biasing power may be controlled so that the ion acceleration energy is between about 20 eV to about 200 eV (e.g., 50 eV to about 150 eV).
(43) In some embodiments, which can be combined with one or more embodiments of the present disclosure, the removal of the third ILD **178** and the etch stop layer **145** may be done by a cyclic process including repetitions of a first etching step (as discussed above with respect to FIGS. **16**A-**16**B) and a second etching step (as discussed above with respect to FIGS. **17**A-**17**B). The cyclic process may use alternating chlorine/oxygen/fluorine-based plasma and chlorine/oxygen/fluorine-based plus argon plasma. For example, the first etching step may use a fluorine/chlorine and oxygen-based plasma and the second plasma etching step may use a fluorine/chlorine and oxygen-based plasma plus argon plasma, or vice versa.
(44) FIG. **17**B-**1** is an enlarged view of a portion **157** of the semiconductor device structure **100** showing the third ILD **178** being etched by the reactive ion beam **155** in accordance with some embodiments. In one embodiment, the reactive ion beam **155** is scanned over the semiconductor device structure **100** along Y direction (e.g., the longitudinal direction of the replacement gate

structures **177**). The reactive ion beam **155** is extracted from the plasma **154** through the extraction aperture **179** and directed to the third ILD **178** and the etch stop layer **145** at an angle β with respect to a direction of a normal line **151** to a top surface **159** of the third ILD **178**. In various embodiments, the angle β is less than about 50 degrees. In some embodiments, the angle β is in a range of about 15 degrees to about 45 degrees, for example about 20 degrees to about 30 degrees. If the angle β is greater than about 50 degrees, the reactive ion beam **155** may land on the sidewall at the top portion **178-1** of the third ILD **178**, which is to be removed during the subsequent CMP process and thus has little or no impact on the sidewall profile of the resulting via contact opening **149′**. In any case, the reactive ion beam **155** is directed at an angle so that the majority of As a result, the purpose of the second etch process is defeated. On the other hand, if the angle β is less than about 15 degrees, the reactive ion beam **155** may hit the exposed conductive features **172** and generate metal residues that may contaminate the transistors and affect the device performance. In some embodiments, the distance D**1** between the extraction parts **153***a*, **153***b* and the top surface **159** of the third ILD **178** may be in a range of about 7 mm to about 14 mm. A skilled artisan in the art may control, for example, the source power, the biasing power, and the distance D**1** between the extraction parts **153***a*, **153***b* and the top surface **159** of the third ILD **178** to adjust the angle β of the ion beam **155**. Additionally or alternatively, the semiconductor device structure **100** may be tilted to alter the direction of impact of the reactive ion beam **155**.

(45) FIG. **18-1** illustrates a top view of a portion of the third ILD **178** showing the profile of the via contact opening **149** prior to the reactive ion beam etch process. FIG. **18-2** illustrates a top view of a portion of the third ILD **178** showing the profile of the via contact openings **149′** after the reactive ion beam etch process in accordance with some embodiments. As discussed above, if the via contact openings **149** have a rounded shape before the second etch process, the via contact openings **149′** may become an elongated circle or oval shape after the reactive ion beam etch process. If the via contact openings **149** have an oval shape, the via contact openings **149′** may become an elongated oval shape after the reactive ion beam etch process. As can be seen, some via contact openings **149**, such as via contact openings **149-1** in a first region of the third ILD **178**, may have a rounded shape, and some via contact openings **149**, such as via contact openings **149-2** in a second region of the third ILD **178**, may have an oval shape. In some embodiments, the via contact openings **149-1** are disposed over a replacement gate structure and the via contact openings **149-2** are disposed over a source/drain epitaxial features. Prior to the reactive ion beam etch process, the via contact openings **149-1** may have a dimension D**2** along the Y-direction and a dimension D**6** along the X-direction, and the via contact openings **149-2** may have a dimension D**3** along the Y-direction and a dimension D**7** along the X-direction. After the reactive ion beam etch process, the via contact openings **149′-1** are extended/elongated along the Y-direction and may have a dimension D**4** greater than the dimension D**2** of the via contact openings **149-1**. The dimension D**8** of the via contact openings **149′-1** extending along the X-direction may be substantially the same or slightly greater than the dimension D**6** of the via contact openings **149-2**. Likewise, the via contact openings **149′-2** are extended/elongated along the Y-direction and may have a dimension D**5** greater than the dimension D**3** of the via contact openings **149-2**. The dimension D**9** of the via contact openings **149′-2** extending along the X-direction may be substantially the same or slightly greater than the dimension D**7** of the via contact openings **149′-2**. In various embodiments, the dimension D**2** and dimension D**6** of the via contact openings **149-1** may be at a ratio (D**2**:D**6**) of about 1:1. The dimension D**4** and dimension D**8** of the via contact openings **149′-1** may be at a ratio (D**4**:D**8**) of about 4:1 to about 8:1, for example about 7:1. Similarly, the dimension D**3** and dimension D**7** of the via contact openings **149-2** may be at a ratio (D**3**:D**7**) of about 2:1 to about 3:1. The dimension D**5** and dimension D**9** of the via contact openings **149′-2** may be at a ratio (D**5**:D**9**) of about 4:1 to about 8:1, for example about 5:1 to about 7:1.

(46) In some embodiments, a cleaning process may be performed between the first etch process and

the second etch process to remove residues from exposed surfaces of the semiconductor device structure **100**. The cleaning process may be any suitable wet cleaning process including, for example, hydrofluoric acid (HF), standard clean 1 (SC1), and ozonated deionized water (DIO.sub.3). In one embodiment, the cleaning process is performed by exposing the semiconductor device structure **100** to HF (1:500 dilution), followed by the DIO.sub.3 rinsing and SC1 cleaning which may be a mixture of deionized (DI) water, ammonia hydroxide (NH.sub.4OH), and hydrogen peroxide (H.sub.2O.sub.2). Other cleaning process, such as an APM process, which includes at least water (H.sub.2O), NH.sub.4OH, and H.sub.2O.sub.2, a HPM process, which includes at least H.sub.2O, H.sub.2O.sub.2, and hydrogen chloride (HCl), a SPM process (also known as piranha clean), which includes at least H.sub.2O.sub.2 and sulfuric acid (H.sub.2SO.sub.4), or any combination thereof, may also be used.
(47) In FIGS. **19**A-**19**B, conductive features **180** are formed in the via contact openings **149**′. The conductive features **180** serve as via contacts for connecting to the epitaxial S/D features **152** through S/D contacts (e.g., conductive features **172**) and connecting to the gate electrode layers **167** through gate contacts (e.g., conductive features **173**). The conductive features **180** may include the same material as the conductive features **172**, **173**, such as one or more of Ru, Mo, Co, Ni. W, Ti, Ta, Cu, Al, TiN and TaN. The conductive feature **180** may be formed by any suitable process, such as PVD, CVD, ALD, electrochemical plating, or other suitable method. Once the conductive features **180** are formed, a planarization process, such as CMP, is performed on the semiconductor device structure **100** until the top surface of the third ILD **178** is exposed.
(48) In FIGS. **20**A-**20**B, an interconnect structure **117** is formed over the semiconductor device structure **100**. In some embodiments, the interconnect structure **117** comprises a plurality of intermetal dielectric (IMD) layers **168***a*-**168***n*, a plurality of conductive features **169***a*-**169***n* embedded in the plurality of IMD layers, and a plurality of etch stop layers **171***a*-**171***n* disposed between ILD and IMD layer (e.g., the third ILD **178** and the first IMD layer **168***a*) and between IMD layers (e.g., the first IMD **168***a* and the second IMD layer (not shown)). The 1 MB layers, the conductive features, and the etch stop layers may repeat until a desired number of the IMD layer **168***n* (e.g., topmost 1 MB layer in the interconnect structure **117**), a desired number of the etch stop layer **171***n*, and a desired number of the conductive features **169***n* (e.g., topmost conductive features in the interconnect structure **117**) embedded in the IMD layer **168***n* is achieved. Conductive features (e.g., conductive vias and conductive lines) may be formed using any suitable formation process (e.g., lithography with etching, damascene, dual damascene, or the like). In some embodiments, the steps for forming the conductive features may include forming openings in the respective dielectric layers, depositing a conductive layer in the openings, and subsequently performing a planarization process, such as a CMP process, to remove excess materials of the conductive material overfilling the openings. The conductive layer may be deposited by CVD, PVD, sputtering, electroplating, electroless plating, or other suitable deposition technique.
(49) The IMD layers **168***a-n* may include or be formed of any suitable dielectric material, such as silicon oxide, a low dielectric constant (low-k) material, or a combination thereof. The low-k material may include fluorinated silica glass (FSG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), polyimide, SiOxCyHz, or SiOxCy, where x, y and z are integers or non-integers, and/or other future developed low-k dielectric materials. The 1 MB layers **168***a-n* may be deposited by a plasma-enhanced CVD (PECVD) process or other suitable deposition technique. The material of the etch stop layers **171***a-n* is chosen such that etch rates of the etch stop layers **171***a-n* are less than etch rates of the 1 MB layers **168***a-n*. In some embodiments, the etch stop layers **171***a-n* may include the same material as the etch stop layer **145** described above. The conductive vias/lines **169***a-n* may include or be formed of any suitable electrically conductive material, such as W, Ru, Co, Cu, Ti, TiN, Ta, TaN, Mo, Ni, or combinations thereof.
(50) The present disclosure provides a method for forming a semiconductor device structure by

increasing contact area between via contacts of the source/drain (S/D) contacts and the gate contacts and the neighboring metal features in the back-end-of-line (BEOL). After formation of the via contact openings using a first etch process, a reactive ion beam etch process is performed to further extend one dimension of the via contact openings so that the dimension of the via contact openings along the Y-direction is at least 7 times greater than the dimension of the via contact openings along the X-direction. The via contact openings can be extended with rounder profile without limitation of lithography. The increased via contact openings allow for better metal gap-filling. Since the via contacts as formed have increased contact area, the contact resistance can be reduced effectively. As a result, high performance of integrated circuits (ICs) can be achieved.

(51) An embodiment is a method for forming a semiconductor device structure. The method includes including forming one or more conductive features in a first interlayer dielectric (ILD), forming an etch stop layer on the first ILD, forming a second ILD over the etch stop layer, forming one or more openings through the second ILD and the etch stop layer to expose a top surface of the one or more first conductive features, wherein the one or more openings are formed by a first etch process in a first process chamber, exposing the one or more openings to a second etch process in a second process chamber so that the shape of the or more openings is elongated, and filling the one or more openings with a conductive material

(52) Another embodiment is a method for forming a semiconductor device structure. The method includes forming a fin structure from a substrate, forming a sacrificial gate structure over a portion of the fin structure, removing portions of the fin structure not covered by the sacrificial gate structure, forming a source/drain feature in regions created as a result of removal of the portions of the fin structure, forming sequentially a contact etch stop layer (CESL) and a first interlayer dielectric (ILD) on the source/drain feature, removing the sacrificial gate structure to expose a portion of the fin structure, forming sequentially a gate dielectric layer and a gate electrode layer on exposed portion of the fin structure, forming a second ILD over the first ILD and the gate electrode layer, forming first contact openings through the second ILD, the first ILD, and the CESL to expose a portion of the source/drain feature, forming second contact openings through the second ILD to expose a portion of the gate electrode layer, forming source/drain contacts and gate contacts by filling the first and second contact openings with a first conductive material, forming sequentially an etch stop layer and a third ILD over the source/drain contacts and the gate contacts, performing a first etch process in a first process chamber to form via contact openings, wherein the via contact openings extend through the third ILD and the etch stop layer to expose the source/drain contacts and the gate contacts, subjecting the via contact openings to a second etch process in a second process chamber, and filling the via contact openings with a second conductive material.

(53) A further embodiment is a method for forming a semiconductor device structure. The method includes forming a first conductive feature in a first interlayer dielectric (ILD) over a gate electrode layer, forming a second conductive feature in the first ILD over a source/drain epitaxial feature, forming a second ILD over the first ILD, forming via openings through the second ILD to expose a top surface of the first conductive feature and a top surface of the second conductive feature, wherein the via openings are etched through a mask by a first etch process, removing the mask, exposing the via openings to a second etch process so that a dimension of the via openings along a first direction is elongated, and filling the via openings with a conductive material.

(54) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that

they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming a semiconductor device structure, comprising: forming one or more conductive features in a first interlayer dielectric (ILD); forming an etch stop layer on the first ILD; forming a second ILD over the etch stop layer; forming one or more openings through the second ILD and the etch stop layer to expose a top surface of the one or more first conductive features, wherein the one or more openings are formed by a first etch process in a first process chamber; exposing the one or more openings to an ion beam of a second etch process in a second process chamber at an angle of about 50 degrees or less with respect to a direction of a normal line to a top surface of the second ILD so that a shape of the or more openings is elongated; and filling the one or more openings with a conductive material.
2. The method of claim 1, wherein the first etch process is a plasma-based etch process and the second etch process is a reactive ion beam etch process.
3. The method of claim 1, wherein the second etch process comprises: forming a plasma in the second process chamber; and directing the ion beam of reactive ions from the plasma to the one or more openings so that a first dimension of the one or more openings is increased to a second dimension after the second etch process.
4. The method of claim 3, further comprising: scanning the ion beam over the one or more openings wherein the beam of reactive ions is directed at an angle of about 50 degrees or less with respect to a direction of a normal line to a top surface of the second ILD.
5. The method of claim 4, wherein the angle is in a range of about 20 degrees to about 30 degrees.
6. The method of claim 3, wherein the second dimension of the one or more openings remain substantially the same after the second etch process.
7. The method of claim 6, wherein the first dimension and the second dimension have a ratio (first dimension: second dimension) in a range of about 1:4 to about 1:8.
8. The method of claim 3, further comprising: applying a bias power to a substrate support on which the semiconductor device structure is disposed.
9. The method of claim 8, wherein the bias power is in a range of about 0.1 kV to about 12 kV.
10. A method for forming a semiconductor device structure, comprising: forming a fin structure from a substrate; forming a sacrificial gate structure over a portion of the fin structure; removing portions of the fin structure not covered by the sacrificial gate structure; forming a source/drain feature in regions created as a result of removal of the portions of the fin structure; forming sequentially a contact etch stop layer (CESL) and a first interlayer dielectric (ILD) on the source/drain feature; removing the sacrificial gate structure to expose the portion of the fin structure; forming sequentially a gate dielectric layer and a gate electrode layer on the exposed portion of the fin structure; forming a second ILD over the first ILD and the gate electrode layer; forming first contact openings through the second ILD, the first ILD, and the CESL to expose a portion of the source/drain feature; forming second contact openings through the second ILD to expose a portion of the gate electrode layer; forming source/drain contacts and gate contacts by filling the first and second contact openings with a first conductive material; forming sequentially an etch stop layer and a third ILD over the source/drain contacts and the gate contacts; performing a first etch process in a first process chamber to form via contact openings, wherein the via contact openings extend through the third ILD and the etch stop layer to expose the source/drain contacts and the gate contacts; subjecting the via contact openings to an ion beam of a second etch process in a second process chamber at an angle of about 50 degrees or less with respect to a direction of a normal line to a top surface of the third ILD so that a dimension of the via openings is elongated; and filling the via contact openings with a second conductive material.

11. The method of claim 10, wherein the second etch process is a reactive ion beam etch process.
12. The method of claim 11, further comprising: applying a biasing power to a substrate support on which the semiconductor device structure is disposed.
13. The method of claim 11, further comprising: scanning the ion beam over the via contact openings.
14. The method of claim 13, wherein the reactive ion beam etch process is performed such that the dimension along the Y-direction is about 7 times the dimension along X-direction.
15. The method of claim 13, wherein the ion beam is extracted from a plasma through an extraction aperture between extraction parts, and a distance between the extraction parts and the top surface of the third ILD is in a range of about 7 mm to about 14 mm.
16. The method of claim 15, wherein the plasma is formed from a fluorine-containing gas, a chlorine-containing gas, an inert gas, an oxygen-containing gas, or combination thereof.
17. The method of claim 10, wherein the etch stop layer has a thickness of about 100 Angstroms to about 300 Angstroms, and the third ILD has a thickness of about 200 Angstroms to about 700 Angstroms.
18. A method for forming a semiconductor device structure, comprising: forming a first conductive feature in a first interlayer dielectric (ILD) over a gate electrode layer; forming a second conductive feature in the first ILD over a source/drain epitaxial feature; forming a second ILD over the first ILD; forming via openings through the second ILD to expose a top surface of the first conductive feature and a top surface of the second conductive feature, wherein the via openings are etched through a mask by a first etch process in a first process chamber; removing the mask; exposing the via openings to a beam of reactive ions at an angle of about 50 degrees or less with respect to a direction of a normal line to a top surface of the second ILD of a second etch process in a second process chamber so that a dimension of the via openings along a first direction is elongated; and filling the via openings with a conductive material.
19. The method of claim 18, wherein the second etch process comprises: applying a bias power to a substrate support on which the semiconductor device structure is disposed; and.
20. The method of claim 19, further comprising: after the first etch process, exposing the via openings to a cleaning process.